ANLP-UniSo at ArGuard 2026 Task A: Multimodal AraBERT–ResNet50 with Knowledge Graph Fusion for Arabic Hateful Meme Detection

Saoussen Ben Chaabane
Faculty of Economics and Management
University of Sfax, Tunisia
ANLP Research Group, MIRACL Laboratory
saoussen.benchaabane@fsegs.usf.tn

Omar Trigui
Higher Institute of Management
University of Sousse, Tunisia
ANLP Research Group, MIRACL Laboratory
omar.trigui@isgs.u-sousse.tn

Maher Jaoua
Faculty of Economics and Management
University of Sfax, Tunisia
ANLP Research Group, MIRACL Laboratory
maher.jaoua@fsegs.usf.tn

Abstract

Detecting hateful content in Arabic memes is challenging due to dialectal variation, sarcasm, cultural references, and the interplay between visual and textual signals. We participate in ArGuard 2026 Track A (Multimodal Hateful Meme Detection) with a multimodal architecture that fuses AraBERT textual representations, ResNet50 visual features, and a lightweight knowledge-graph (KG) ontology vector derived from domain-specific Arabic keywords. The model jointly optimizes binary (Hateful / Not Hateful) and multi-label fine-grained classification through a gated fusion layer with strong regularization (dropout 0.45, weight decay 0.05, label smoothing 0.1). Gradual unfreezing of AraBERT layers, weighted sampling, and Arabic-specific normalization further improve robustness. On the official development set the system reaches a binary macro-F1 of 0.710; on the blind test set it obtains binary macro-F1 0.717. We analyse the contribution of each modality and discuss remaining challenges of fine-grained multi-label prediction under severe class imbalance.

1 Introduction

Memes have become a prevalent vehicle for online expression and, unfortunately, for the spread of hate speech. In Arabic-speaking communities the problem is amplified by dialectal diversity, code-switching, sarcasm, and culturally specific visual tropes (Alam et al., 2026). ArGuard 2026 Track A addresses this need with two subtasks:

- **A1 (Binary)**: Classify an Arabic meme (image + OCR text) as Hateful or Not Hateful.
- **A2 (Fine-grained)**: Predict the applicable multi-label categories from a unified taxonomy that covers both hateful subtypes (Mocking, Incitement, Dehumanization, Slurs, Contempt, Inferiority, Exclusion, ...) and non-hateful categories (Humor, Sarcasm, Other).

We participate with a single multimodal system based on AraBERT (Antoun et al., 2020) for text, ResNet50 (He et al., 2016) for images, and a knowledge-graph ontology that injects lexical cues for hate-related concepts. Our contributions are: (i) a gated three-way fusion of text, image and KG features; (ii) strong regularization and gradual fine-tuning tailored to the small training set (3 500 memes); and (iii) an explicit multi-label head with class-weighted BCE that handles the extreme imbalance of fine-grained labels.

2 Related Work

Research on Arabic hate speech detection has progressively shifted from conventional machine learning methods toward pretrained language models that better capture contextual and dialectal variation. Early approaches commonly relied on lexical representations such as TF-IDF with classifiers including SVM and Logistic Regression, while recent studies increasingly employ Arabic-specific transformers such as AraBERT (Antoun et al., 2020) and MARBERT (Abdul-Mageed et al., 2021). Shared tasks such as OSACT have also contributed to the development of Arabic offensive-language classifiers (Mubarak et al., 2020). More recently, MAHED 2025 extended this research to hate-related content and multimodal Arabic memes (Zaghouani et al., 2025).

A recent MAHED 2025 system by (El Abed et al., 2025) further illustrates the use of contextual pretrained representations for Arabic harmful-content classification. The ANLP-UniSo system used XLM-RoBERTa representations combined

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 448–452
October 28-29, 2026 ©2026 Association for Computational Linguistics

with Logistic Regression and LSTM, while applying SMOTE to address class imbalance. The system achieved a macro score of 0.60 (El Abed et al., 2025). These results highlight the relevance of pretrained representations and imbalance-aware learning in limited-data Arabic settings.

Dialectal variation remains a major challenge for Arabic hate-speech classification. Differences in vocabulary and contextual meaning across dialects can affect classification performance. Abdelsamie et al. (2026) addressed this issue through a multi-task transformer framework covering Egyptian, Saudi, Levantine, Gulf, and Algerian Arabic. By learning shared and dialect-specific representations, their approach improved hate-speech detection across the five dialects and demonstrated the importance of explicitly modelling dialectal characteristics (Abdelsamie et al., 2026).

Recent research has also shifted from text-only hate-speech detection toward multimodal Arabic meme analysis, as harmful meaning may emerge from the interaction between textual and visual signals. ArAIEval 2024 established a dedicated multimodal Arabic meme setting for propaganda detection (Hasanain et al., 2024). Within this benchmark, Alhabashi et al. (2024) combined MARBERT and ResNet50 through a fusion layer for Arabic meme classification (Alhabashi et al., 2024). Similarly, the ArMeme resource introduced a large-scale benchmark for propagandistic Arabic memes and evaluated several visual architectures, including ResNet50 (Alam et al., 2024). These studies establish text–image fusion as an effective paradigm for Arabic meme understanding.

Recent work has further targeted hateful rather than solely propagandistic meme content. AHA-Memes introduces a fine-grained Arabic hateful-meme benchmark with 5K manually annotated memes and multi-label hate categories (Kmainasi et al., 2026). The benchmark evaluates text-only, image-only, and multimodal approaches and shows that fine-grained hateful-meme classification remains challenging even when both modalities are available (Kmainasi et al., 2026). This highlights the need for models capable of handling multiple interacting hate categories rather than only coarse binary decisions.

Despite these advances, existing Arabic multimodal approaches predominantly rely on learned textual and visual representations, with relatively limited use of explicit structured knowledge. Knowledge graphs and lexical ontologies can complement learned representations with domain-specific concepts, particularly for rare or dialect-specific hate categories. However, their integration into multimodal Arabic hateful-meme detection remains comparatively underexplored. ArGuard 2026 provides an appropriate setting for addressing this limitation through its binary and fine-grained multi-label formulation (Alam et al., 2026). Building on previous work in Arabic transformers, dialect-aware modelling, and multimodal meme analysis, our system combines AraBERT and ResNet50 with a lightweight domain lexical-ontology to provide complementary textual, visual, and semantic cues for Arabic hateful-meme classification.

3 Data and Preprocessing

We use the official ArGuard Task A data (Alam et al., 2026) (QCRI/ArGuard-Task1 on Hugging Face). The splits contain 3 500 (train), 500 (dev), 500 (dev_test) and 500 (blind test) memes. Binary label distribution on train is 1 324 Hateful / 2 176 Not Hateful (≈ 37.8% hateful). Fine-grained labels are heavily skewed: Sarcasm (934), Humor (863) and Mocking (706) dominate, while Exclusion, Inferiority and several other classes appear fewer than 60 times.

Figure 1 shows a training example labeled *Hateful* with fine-grained tags Dehumanization and Mocking: the image (Tom & Jerry carrying a fridge) interacts with the overlaid Arabic text to convey the hateful meaning. In many cases the hateful intent emerges only from the interaction between the image and the overlaid Arabic text rather than from either modality alone.



Figure 1: Example training meme labeled Hateful (Dehumanization, Mocking).

Text preprocessing consists of (i) removal of URLs, mentions and hashtags, (ii) Arabic normalization (tashkeel/tatweel removal, alef/yeh/waw unification, reduction of character repetitions), and (iii) a small dialect-to-MSA replacement dictionary covering common Maghrebi and Egyptian forms.

Images are resized to 224×224 and normalized with ImageNet statistics; training uses RandomResizedCrop, RandomHorizontalFlip, ColorJitter, RandomAffine and RandomErasing. No external labeled datasets were used. The official labeled development set is employed for both checkpoint selection and hyper-parameter tuning.

4 System Architecture and Training

4.1 Model

The core model is a three-stream encoder followed by gated fusion:

- **Text stream**: AraBERT (`aubmindlab/bert-base-arabertv2`). Embeddings and the first 8 encoder layers are frozen at the start of training; the last 4 layers are gradually unfrozen after epoch 3.
- **Image stream**: ResNet50 (ImageNet pretrained). All layers are frozen except layer 4, which is fine-tuned.
- **Lexical-ontology stream**: We construct a lightweight domain ontology of 15 fine-grained hate-related categories as a NetworkX graph, with each category linked to curated Arabic keywords. Manually defined relations between categories (e.g., Slurs $\leftrightarrow$ Dehumanization) enable limited score propagation. Each meme is represented by a 15-dimensional sparse, L1-normalized ontology feature vector. Unlike a general-purpose knowledge graph, this ontology is manually curated and specific to the target hate-speech taxonomy.

Each stream is projected to a common fusion dimension of 128 by a linear layer + LayerNorm + GELU + Dropout. The three projected vectors are concatenated and modulated by a learned sigmoid gate. A subsequent MLP with BatchNorm and heavy dropout (0.45) yields a 64-dimensional representation that feeds two heads: a binary linear head (2 classes) and a multi-label linear head (15 classes). Figure 2 provides an overview of the full system. Table 1 summarizes the main components and training choices.

4.2 Objectives and Optimization

Binary classification uses cross-entropy with label smoothing 0.1. Fine-grained multi-label classification uses BCEWithLogitsLoss with inverse-frequency positive weights (clipped at 50). We train with AdamW (backbone lr = 1×10^{-5}, head lr = 5×10^{-4}, weight decay = 0.05), mixed-precision, gradient clipping, and a linear warm-up schedule. Batch size is 16, maximum sequence length 128, and training runs for a maximum of 8 epochs with early stopping (patience 3) on development binary macro-F1. A WeightedRandomSampler balances the binary classes inside each epoch. The best dev Macro-F1 increases from 0.7028 before unfreezing to 0.7096 after unfreezing the last four AraBERT layers, after which performance degrades, indicating overfitting and triggering early stopping.

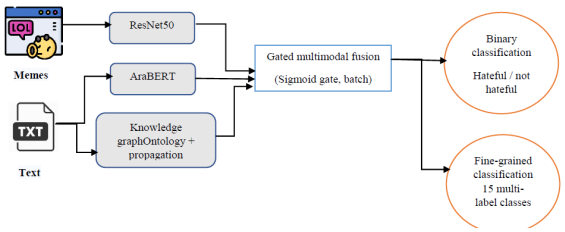


Figure 2: Overview of the multimodal architecture.

Component	Choice
Text encoder	AraBERT (`arabertv2`)
Image encoder	ResNet50 (layer 4 fine-tuned)
Knowledge stream	15-class keyword ontology + propagation
Fusion	Gated concat (sigmoid, BN, dim 128)
Binary head	Softmax, 2 classes
Fine-grained head	Sigmoid, 15 multi-label classes
Regularization	Dropout 0.45, WD 0.05, label smoothing 0.1
Sampling	WeightedRandomSampler (binary)
Image aug.	Crop, flip, ColorJitter, Affine, Erasing
Schedule	Unfreeze after epoch 3, early stopping

Table 1: Summary of system components and training choices.

5 Experimental Results

Table 2 reports development and official blind-test metrics for the two subtasks. The model achieves binary Macro-F1 scores of 0.710 on dev and 0.717 accuracy 0.760), on the official test set. Fine-grained Macro-F1 is 0.183 on dev and 0.320 on test, with test Micro-F1 of 0.411 and subset accuracy of 0.006.

Figure 3 shows the binary confusion matrices on the development set (absolute and row-normalized). The model is better at recognizing Not-Hateful memes (recall ≈ 0.82) than Hateful ones (recall

Binary (A1)		Fine-grained (A2)	
Split	**Macro-F1**	**Split**	**Macro-F1**
Dev	0.710	Dev	0.183
Test	0.717	Test	0.320

Table 2: Official results for Subtasks A1 (binary; test Acc. 0.760) and A2 (fine-grained; test Micro-F1 0.411, Subset Acc. 0.006).

≈ 0.59), a pattern consistent with the class imbalance and the subtlety of many hateful memes that rely on visual irony. Figure 4 shows the training dynamics: the best development macro-F1 is reached around epoch 3–4, after which the train–dev gap widens and early stopping triggers.

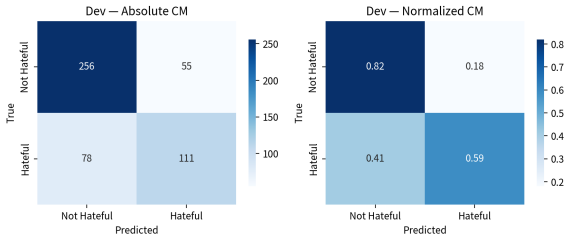


Figure 3: Binary confusion matrices on the development set (left: absolute counts; right: row-normalized).

Note that Figure 3 is computed on *dev*, the only split with available gold labels. The `dev_test` and official blind test splits have labels withheld by the organizers; test performance is therefore obtained only from the official leaderboard (Table 2), and no test confusion matrix can be computed..

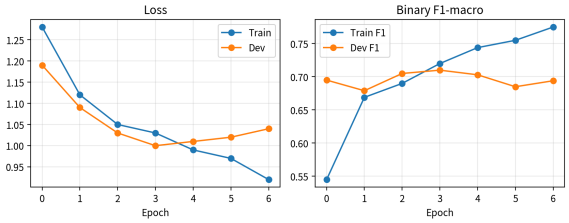


Figure 4: Training curves for loss (left) and binary macro-F1 (right).

6 Analysis and Discussion

Table 3 reports a training-time ablation on the development set. Text and image modalities are individually informative (Macro-F1 0.6692 and 0.6576, respectively), while their fusion improves performance to 0.7107. Adding the KG stream results in 0.7096, indicating that the ontology does not provide an additional binary-performance gain on this split. This suggests that the text and image streams dominate the binary decision, while the ontology may be more useful for lexical cues in fine-grained categories.

Setting	Best Epoch	Dev Macro-F1
Text only	4	0.6692
Image only	3	0.6576
Text + Image (no KG)	3	0.7107
Text + Image + KG (full)	4	0.7096

Table 3: Ablation: binary Macro-F1 on *dev*.

The fine-grained task remains challenging because several rare classes have fewer than 60 positive training examples, while semantically overlapping categories, such as Mocking vs. Sarcasm and Contempt vs. Insults, make multi-label macro-F1 particularly difficult. Limited dialectal coverage also contributes to errors on short or highly dialectal captions.

7 Conclusion

We presented a multimodal system for Arabic hateful-meme classification combining AraBERT, ResNet50, and a lightweight lexical ontology. The system achieves a binary Macro-F1 of 0.717 on the official test set, while fine-grained multi-label prediction remains challenging under severe class imbalance. Future work will investigate larger vision-language models, hierarchical multi-label objectives, and dialect-aware augmentation.

8 Limitations

The system relies exclusively on the official ArGuard data and was trained on a single GPU. Using the development set for model selection introduces potential selection bias, while limited dialectal augmentation and long-tailed fine-grained labels remain important limitations.

9 Ethical Considerations

Hate-speech classifiers may be misused for over-censorship or may fail to protect vulnerable users when harmful content is missed. We used only the official shared-task data for research purposes and recommend human oversight and continuous monitoring when deploying such systems.

Acknowledgments

We thank the ArGuard 2026 organizers for preparing the shared task and releasing the dataset.

References

Mahmoud Mohamed Abdelsamie, Shahira Shaaban Azab, and Hesham A. Hefny. 2026. The dialects gap: A multi-task learning approach for enhancing hate speech detection in arabic dialects. *Expert Systems with Applications*, 295:128584.

Muhammad Abdul-Mageed, AbdelRahim Elmadany, and El Moatez Billah Nagoudi. 2021. ARBERT & MARBERT: Deep bidirectional transformers for Arabic. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 7088–7105.

Firoj Alam, Md. Rafiul Biswas, Mohamed Bayan Kmainasi, Ali Ezzat Shahroor, Hamdy Mubarak, Georgios Mikros, Abul Hasnat, and Wajdi Zaghouani. 2026. ArGuard shared task: Harmful content detection in Arabic memes and LLM prompts. In *Proceedings of The Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Firoj Alam, Abul Hasnat, Fatema Ahmad, Md. Arid Hasan, and Maram Hasanain. 2024. ArMeme: Propagandistic content in Arabic memes. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 21071–21090.

Yasser Alhabashi, Abdullah Alharbi, Samar Ahmad, Serry Sibaee, Omer Nacar, Lahouari Ghouti, and Anis Koubaa. 2024. ASOS at ArAIEval shared task: Integrating text and image embeddings for multimodal propaganda detection in Arabic memes. In *Proceedings of the Second Arabic Natural Language Processing Conference*, pages 473–477, Bangkok, Thailand. Association for Computational Linguistics.

Wissam Antoun, Fady Baly, and Hazem Hajj. 2020. AraBERT: Transformer-based model for Arabic language understanding. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools*, pages 9–15.

Yasmine El Abed, Mariem Ben Arbia, Saoussen Ben Chaabane, and Omar Trigui. 2025. ANLP-UniSo at MAHED shared task: Detection of hate and hope speech in Arabic social media based on XLM-RoBERTa and logistic regression. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, pages 639–644, Suzhou, China. Association for Computational Linguistics.

Maram Hasanain et al. 2024. ArAIEval shared task: Propagandistic techniques detection in unimodal and multimodal Arabic content. In *Proceedings of the Second Arabic Natural Language Processing Conference*. Association for Computational Linguistics.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.

Mohamed Bayan Kmainasi, Ali Ezzat Shahroor, Abul Hasnat, Md. Rafiul Biswas, Wajdi Zaghouani, and Firoj Alam. 2026. Aha-memes: A fine-grained multimodal benchmark for understanding hate in arabic memes. *Preprint*, arXiv:2607.27393.

Hamdy Mubarak, Kareem Darwish, Walid Magdy, Tamer Elsayed, and Hend Al-Khalifa. 2020. OSACT4 shared task on offensive language detection in Arabic. In *Proceedings of the 4th Workshop on Open-Source Arabic Corpora and Processing Tools*, pages 66–70.

Wajdi Zaghouani, Md. Rafiul Biswas, Mabrouka Bessghaier, Shimaa Ibrahim, Georgios Mikros, Abul Hasnat, and Firoj Alam. 2025. MAHED shared task: Multimodal detection of hope and hate emotions in Arabic content. In *Proceedings of The Third Arabic Natural Language Processing Conference: Shared Tasks*, Suzhou, China. Association for Computational Linguistics.